

University of Prishtina “Hasan PRISHTINA”



Faculty of Electrical and Computer Engineering
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Course: Data Preparation and Visualization

Week VIII

Importance of data volume, similarities and non-similarities - I

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Similarity and Dissimilarity

➤ Similarity:

- Numerical measure of how alike two data objects are;
- Is higher when objects are more alike;
- Often falls in the range $[0,1]$.

➤ Dissimilarity:

- Numerical measure of how different are two data objects;
- Lower when objects are more alike;
- Minimum dissimilarity is often 0;
- Upper limit varies.

➤ Proximity refers to a similarity or dissimilarity.

Similarity / Dissimilarity for Simple Attributes

p and q are the attribute values for two data objects.

Attribute Type	Dissimilarity	Similarity
Nominal	$d = \begin{cases} 0 & \text{if } p = q \\ 1 & \text{if } p \neq q \end{cases}$	$s = \begin{cases} 1 & \text{if } p = q \\ 0 & \text{if } p \neq q \end{cases}$
Ordinal	$d = \frac{ p-q }{n-1}$ (values mapped to integers 0 to $n-1$, where n is the number of values)	$s = 1 - \frac{ p-q }{n-1}$
Interval or Ratio	$d = p - q $	$s = -d,$

Table 5.1. Similarity and dissimilarity for simple attributes

Euclidean Distance

- The Euclidean distance, d , between two points, x and y , in one, two, three, or higher dimensional space, is given by the following familiar formula:

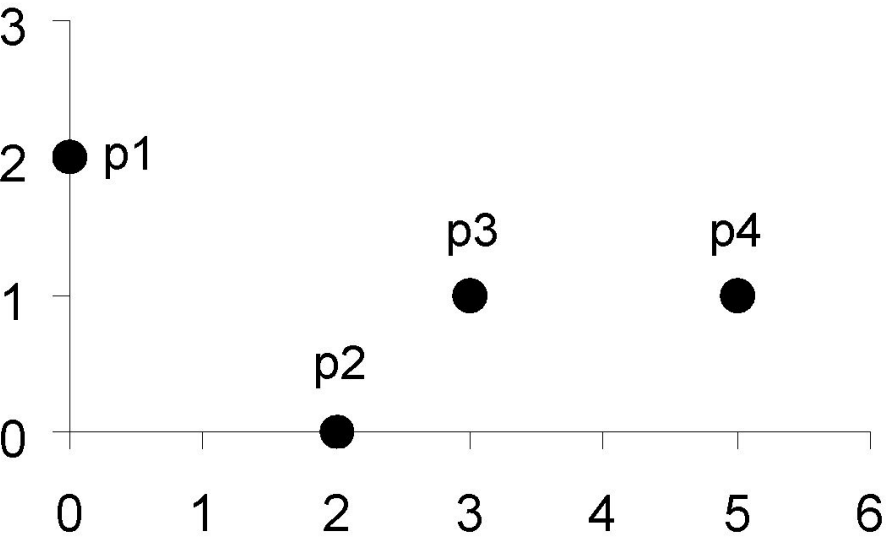
$$\mathbf{dist} = \sqrt{\sum_{k=1}^n (\mathbf{p}_k - \mathbf{q}_k)^2}$$

- Where n is the number of dimensions (attributes) and p_k and q_k are, respectively, the k^{th} attributes (components) or data objects p and q .

$$\mathbf{d} (\mathbf{p}_i, \mathbf{p}_j) = \sqrt{(\mathbf{x}_i - \mathbf{x}_j)^2 + (\mathbf{y}_i - \mathbf{y}_j)^2}$$

- Standardization is necessary, if scales differ.

Euclidean Distance

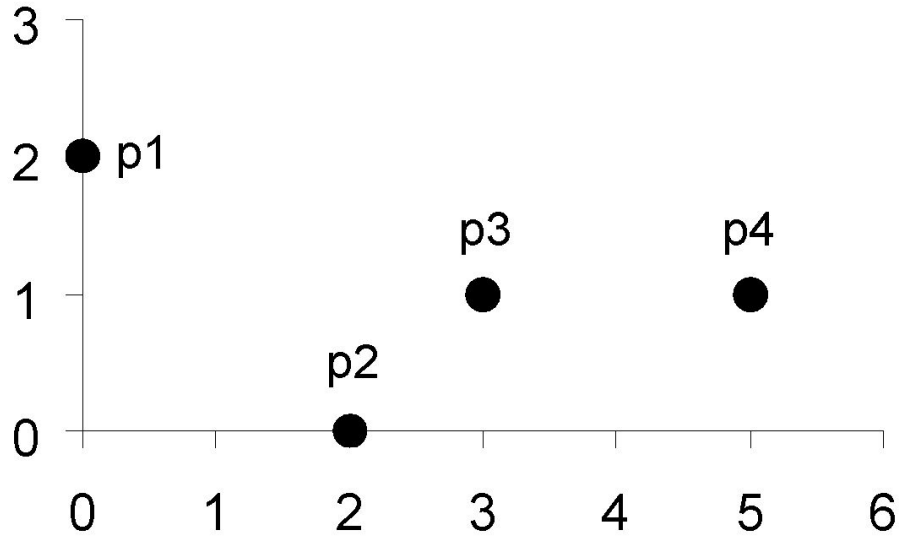


point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

Distance Matrix

	p1	p2	p3	p4
p1				
p2				
p3				
p4				

Euclidean Distance



point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

Distance Matrix

	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

Minkowski Distance

➤ Minkowski Distance is a generalization of Euclidean Distance:

$$\text{dist} = \left(\sum_{k=1}^n |p_k - q_k|^r \right)^{\frac{1}{r}}$$

Where r is a parameter, n is the number of dimensions (attributes) and p_k and q_k are, respectively, the k^{th} attributes (components) or data objects p and q .

Minkowski Distance: Example

➤ $r = 1$. City block (Manhattan, taxicab, L_1 norm) distance:

- A common example of this is the Hamming distance, which is just the number of bits that are different between two binary vectors.

➤ $r = 2$. Euclidean distance;

➤ $r \rightarrow \infty$. “supremum” ($L_{\max}$ norm, L_{∞} norm) distance:

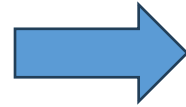
- This is the maximum difference between any component of the vectors.

$$d_5(P, Q) = \max\{|x_1 - x_2|, |y_1 - y_2|\}$$

➤ *Do not confuse r with n , i.e., all these distances are defined for all numbers of dimensions.*

Minkowski Distance: Example

Manhattan Distance



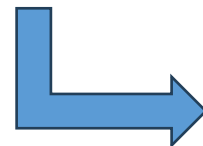
L1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1



L2	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

Distance Matrix



L_{∞}	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

Common Properties of a Distance

➤ Distances, such as the Euclidean distance, have some well-known properties:

1. $d(p, q) \geq 0$ for all p and q and $d(p, q) = 0$ only if:
 $p = q$. (Positive definiteness);
2. $d(p, q) = d(q, p)$ for all p and q . (Symmetry);
3. $d(p, r) \leq d(p, q) + d(q, r)$ for all points p, q , and r .
(Triangle Inequality);

where $d(p, q)$ is the distance (dissimilarity) between points (data objects), p and q .

Common Properties of a Similarity

➤ Similarities, also have some well-known properties:

- $s(p, q) = 1$ (or maximum similarity) only if $p = q$
- $s(p, q) = s(q, p)$ for all p and q . (Symmetry)

where $s(p, q)$ is the similarity between points (data objects), p and q .

Similarity between Binary Vectors

- Common situation is that objects, p and q, have only binary attributes;
- Compute similarities using the following quantities:
 - M_{01} = the number of attributes where p was 0 and q was 1
 - M_{10} = the number of attributes where p was 1 and q was 0
 - M_{00} = the number of attributes where p was 0 and q was 0
 - M_{11} = the number of attributes where p was 1 and q was 1
- **Simple Matching and Jaccard Coefficients (SMC & JC):**
 - SMC = number of matches / number of attributes
 - $$= (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00})$$
 - J = number of 11 matches / number of not-both-zero attributes' values
 - $$= (M_{11}) / (M_{01} + M_{10} + M_{11})$$

SMC versus Jaccard: Example

$p = 1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0$

$q = 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 1$

$M_{01} = 2$ (the number of attributes where p was 0 and q was 1)

$M_{10} = 1$ (the number of attributes where p was 1 and q was 0)

$M_{00} = 7$ (the number of attributes where p was 0 and q was 0)

$M_{11} = 0$ (the number of attributes where p was 1 and q was 1)

$$SMC = (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00}) = ?$$

$$J = (M_{11}) / (M_{01} + M_{10} + M_{11}) = ?$$

SMC versus Jaccard: Example

p = 1 0 0 0 0 0 0 0 0 0

q = 0 0 0 0 0 0 1 0 0 1

$M_{01} = 2$ (the number of attributes where p was 0 and q was 1)

$M_{10} = 1$ (the number of attributes where p was 1 and q was 0)

$M_{00} = 7$ (the number of attributes where p was 0 and q was 0)

$M_{11} = 0$ (the number of attributes where p was 1 and q was 1)

$$SMC = (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00}) = (0+7) / (2+1+0+7) = 0.7$$

$$J = (M_{11}) / (M_{01} + M_{10} + M_{11}) = 0 / (2 + 1 + 0) = 0$$

SMC versus Jaccard – Class Exercise

p = 1 1 0 1 0 1 0 1 1 0

q = 0 1 0 1 0 0 1 0 0 1

$M_{01} = 2$ (the number of attributes where p was 0 and q was 1)

$M_{10} = 4$ (the number of attributes where p was 1 and q was 0)

$M_{00} = 2$ (the number of attributes where p was 0 and q was 0)

$M_{11} = 2$ (the number of attributes where p was 1 and q was 1)

$$SMC = (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00}) ?$$

$$J = (M_{11}) / (M_{01} + M_{10} + M_{11}) = ?$$

SMC versus Jaccard – Class Exercise

p = 1 1 0 1 0 1 0 1 1 0

q = 0 1 0 1 0 0 1 0 0 1

$M_{01} = 2$ (the number of attributes where p was 0 and q was 1)

$M_{10} = 4$ (the number of attributes where p was 1 and q was 0)

$M_{00} = 2$ (the number of attributes where p was 0 and q was 0)

$M_{11} = 2$ (the number of attributes where p was 1 and q was 1)

$$SMC = (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00}) = 0.4$$

$$J = (M_{11}) / (M_{01} + M_{10} + M_{11}) = 0.25$$

Questions?

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